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A novel bearing imbalance Fault-diagnosis method based on a Wasserstein conditional generative adversarial network

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ABSTRACT

In practical industrial applications, rolling bearings are in normal operation most of the time, and the inadequacy of the fault data makes the data itself exhibit unbalanced characteristics. The imbalance of data is difficult to meet the training demand of intelligent networks, which in turn causes its low recognition ability. To solve this problem, this paper proposes a new unbalanced fault diagnosis framework, the Wasserstein conditional generation adversarial network, based on hierarchical feature matching. Unlike traditional generative frameworks, which achieve the overall alignment between the generated distribution and the true distribution by adversarial only. The proposed framework unites Wasserstein loss and hierarchical feature matching loss, and constrain the data generation characteristics from the perspective of global and class-specific to improve the validity of the data. Experiments containing real bearing faults demonstrate the generalization performance of the proposed method, and the results show that the proposed method requires only a small number of 10 samples to reach 92% correct diagnosis, which provides a feasible tool for solving the current industrial data imbalance problem.

1. Introduction

Modern machinery and equipment are widely used in various fields such as construction, aviation, electric power, and metallurgy. Machine failure can cause huge economic losses and pose a serious threat to workers [1–3]. As the core component of rotating machinery, rolling bearings play a vital role in the normal operation of the equipment. Therefore, effective fault-diagnosis technology improves the safety and reliability of the machine and prevents possible damage [4–5].

With the continuous development of data mining and information technology, fault diagnosis has experienced a process from manual-experience diagnosis [6–8] to intelligent-learning diagnosis [9–10]. Li et al. [11] proposed a multi-scale local feature learning method based on back propagation neural network (BPNN) for rolling bearing fault diagnosis. Eren et al. [12] proposed a compact adaptive one-dimensional convolutional neural network, which could intelligently extract information and make decisions in an end-to-end form. Li et al. [13] proposed a rolling-bearing fault-diagnosis method based on deep learning, in which an attention mechanism was introduced to assist the deep-network positioning information segment to extract the discriminant features of input information. Khatir et al. [14] proposed an effective crack recognition method, which used Jaya algorithm to optimize the most important parameters of artificial neural network and improve the effectiveness of the algorithm. Xue et al. [15] established a two-stream feature fusion convolutional neural network to extract depth features from one-dimensional CNN and two-dimensional CNN parallel multi-channel structures to obtain reliable bearing diagnostic results. Hoang et al. [16] provided a systematic overview of bearing fault diagnosis based on deep learning.

However, to ensure the effectiveness of fault diagnosis models, complete labeled training data is necessary. Otherwise, the inductive bias of the network will affect its generalization ability, in turn [17]. Unfortunately, in practice, the quality of industrial datasets is difficult to guarantee. Most machines work in a normal state, with failures accounting for only a tiny fraction of their life cycle. Besides, the lack of acquisition equipment further increases the difficulty of faulty data acquisition. Therefore, the data of the real industry will show unbalanced characteristics, as shown in Fig. 1 with an excess of normal data and a lack of fault data. This makes the dataset characterized by a long-tailed distribution, which prevents deep learning models from being adequately trained and leads to low classification rates in industrial applications [18–20].

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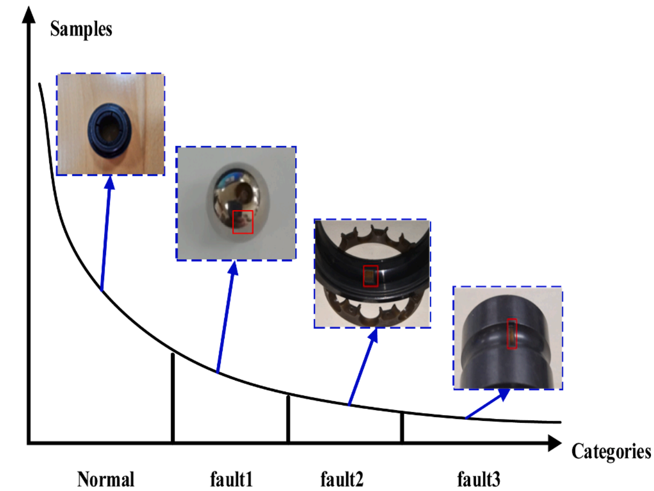

Fig. 1. The long-tailed distribution.

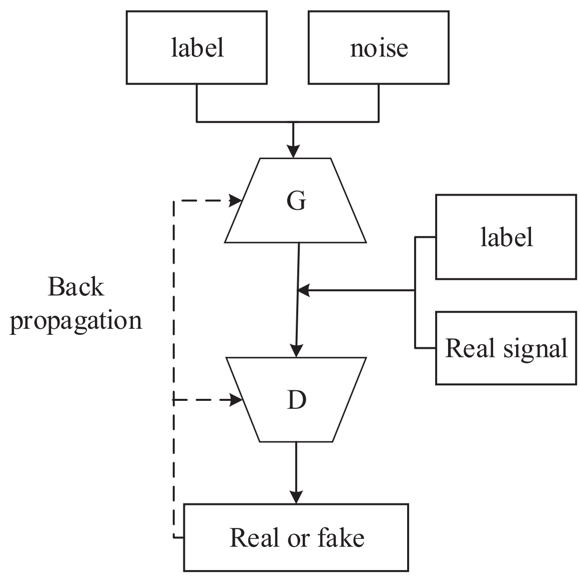

Fig. 2. The basic structure of CGAN.

In dealing with the problem of data imbalance, the data expansion approach adds diversity to the data and essentially solves the data imbalance problem, thus receiving wide attention. Yan et al [21]. proposed a fault-diagnosis method for mechanical equipment, which used a generative adversarial network to generate augmented data to optimize the structure of the original data set. Xu et al [22]. proposed semi-supervised conditional generative adversance network and spectrum normalization, which solved the imbalance problem by data expansion based on time-frequency matrix obtained by wavelet transform. Han et al [23]. combined the excellent feature processing ability of convolutional neural network with the excellent generalization ability of support vector machine (SVM), and proposed CNN-SVM for bearing fault diagnosis. Wang et al [24]. proposed enhanced generative adversarial networks (E-GAN), which generate fault samples through GAN. Aiming at the expanded data, the authors proposed an algorithm combining CNN and K-means to enhance the generalization of the model.

The above studies demonstrate that the GAN-based sample expansion method can effectively solve the data imbalance problem in fault diagnosis. However, there are still some challenges in using GANs, such as the training of GANs is not smooth and may converge to local minima [25]. Besides, the leading training goal of GAN is the indistinguishability of the generated distribution from the true distribution, which focuses on the overall alignment between distributions and ignores the details of specific classes within the distribution. The overall minimization of training loss will force the network to pay more attention to the classification accuracy of normal signals that occupy most of the data, and ignore the mining of fault signal characteristics. These issues reduce the authenticity and validity of the traditional GAN-based generated data, which in turn makes it difficult to guarantee the generalization performance of subsequent classification networks.

To solve the above-mentioned problem, this study proposed a new data-generation framework named the Wasserstein conditional generative adversarial network with hierarchical feature matching (WCGAN-HFM). WCGAN-HFM compensates for the class imbalance deficiencies of the original dataset through extended fault samples, enabling the model to fully exploit the relationships between fault patterns. To provide a stable and effective gradient information for network training, this paper introduced the Wasserstein distance regularization term. Meanwhile, to regularize the generation characteristics of the model from a global and local perspective, this paper proposes a hierarchical feature matching loss, which acts on a specific category in the data to constrain the similarity of local categories in the network's deep variables. In general, the contributions of this paper are as follows:

1) An WCGAN-HFM framework is proposed for handling sample imbalance.
2) Wasserstein loss is introduced to stabilize the training process.
3) Hierarchical feature matching is proposed to regularize the data generation process
4) Simulation and real fault bearing experiments prove the superiority of WCGAN-HFM

The rest of this article is organized as follows. Section II reviews GAN. Section III gives the proposed WCGAN-HFM. Section IV confirms the validity of the WCGAN-HFM with experimental data. Finally, Section V draws the conclusion.

2. A brief review of GAN and related works

As an emerging deep-learning model, generative adversarial networks (GANs) have achieved significant application results in speech recognition, image restoration, and other fields [26–28]. Compared with traditional generative-learning algorithms, the GAN does not depend on the prior setting requirements for data distribution, which gives the GAN a great "free" fitting ability. In theory, being able to fit the data distribution consistently with the original distribution is one of the advantages of the GAN. Thus, it has gradually been applied to deal with small samples, data imbalance, and other issues (for data-generation expansion).

However, owing to the complete abandonment of the a priori setting, when the distribution scale reaches a certain order of magnitude, the GAN itself will not be able to control the direction of the generated results. In 2014, Mirza et al [29]. introduced the category label as a conditional constraint into the generator of a GAN named the conditional generative adversarial network (CGAN). By embedding label information at the input of the generator, the CGAN has the ability to generate specific types of data in a targeted manner. The basic structure of the CGAN is shown in Fig. 2.

The idea of a CGAN is to introduce conditional constraints (such as category labels) into the model, so that the generation process has directionality, and this directionality is continuously trained in the process of error back propagation [29]. The whole process can be regarded as a two-player game between the generator G and the discriminator D, which introduces conditional-probability information:

$$\min_G \max_D V_{CGAN}(D,G) = E_{x\sim P_{data}(x)}[logD(x|y)] + E_{z\sim P_G(z)}[log(1-D(G(z|y)))] \tag{1}$$

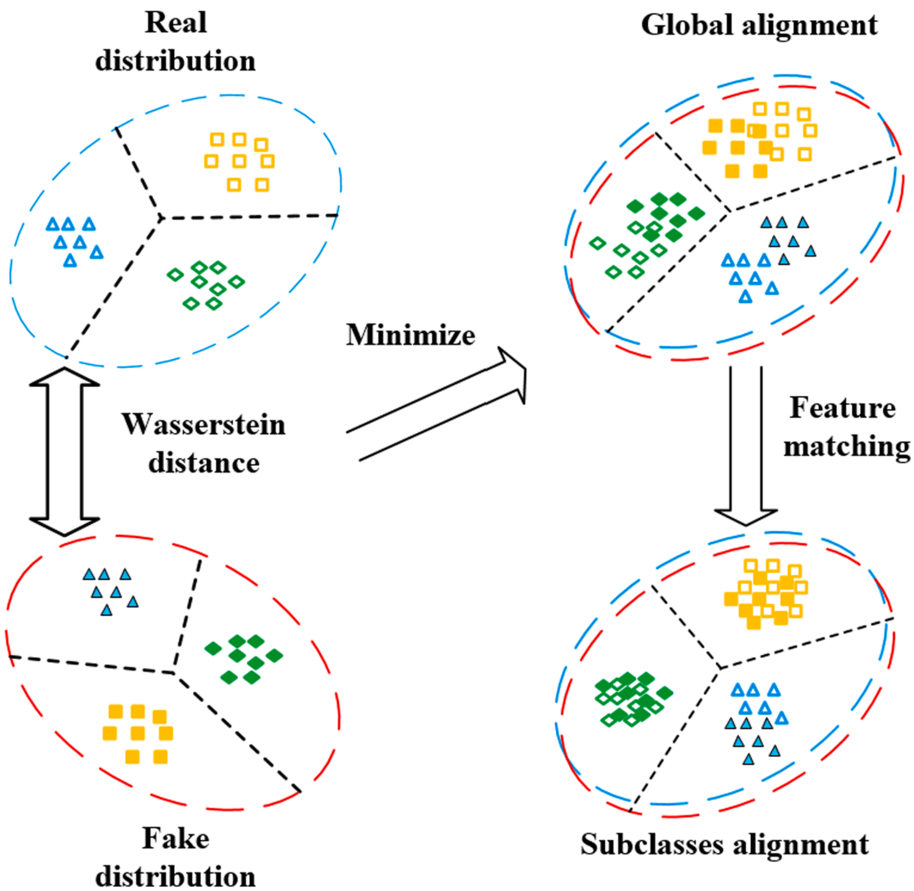

Fig. 3. Wasserstein distance and HFM.

where y is the given conditional information, P_{data} is the real distribution, z is the sampling signal, and $P_G(z)$ is the generated distribution.

The discriminator D tries to maximize the loss function so that it can discriminate the authenticity of the data as much as possible. The generator G tries to minimize the loss function so that the generated data can confuse the discriminator. In the training process, D and G play games with each other until reaching the Nash equilibrium point. During training, a traditional CGAN generally uses Jensen-Shannon (JS) divergence to characterize the overall loss function:

$$\min_G \max_D V_{CGAN}(D,G) = -2\log 2 + \min_G [2D_{JS}(P_{data}||P_G)] \tag{2}$$

where D_{JS} is the JS divergence.

For the generator G, the ideal state is that the real-data distribution and the sampling distribution can be close enough to optimize the JS-divergence distance to the smallest value, so that the generator can generate data with the same distribution characteristics as the real sample. However, minimizing the JS-divergence distance has a prerequisite in which there needs to be an overlap between the generated distribution and the true distribution. In the actual training process, the generator's ability to optimize the generation of samples using feedback training is insufficient, and there may be little or no overlap between the generator's sampling distribution and the real distribution. This will make the JS divergence in Equation (2) meaningless, resulting in the disappearance of the gradient of the loss function and the emergence of the phenomenon of "modal collapse".

3. The proposed WCGAN-HFM

3.1. Wcgan

The Wasserstein distance is a measure that reflects the degree of difference between probability distributions [30], that is, the shortest path required when one data distribution is continuously fitted to approximate another distribution:

$$W_{|p,q|} = \inf_{\gamma\sim\prod(p,q)} E_{(x,y)\sim\gamma}||x-y|| \tag{3}$$

where $\prod(p,q)$ is the set of all possible joint distributions between distribution p and distribution q, $(x,y)\sim\gamma$ is a single sample data of joint distribution γ, and $E_{(x,y)\sim\gamma}||x-y|$ is the expected value of the sample distance of x and y. The Wasserstein distance is the lower bound of the distance in all possible joint distributions $\prod(p,q)$.

Compared with the JS divergence, the calculation of the Wasserstein distance should be smoother, because it also has a good gradient performance for the distance-measurement distribution in the case of no overlap. Using the Wasserstein distance instead of the JS divergence as the metric used to measure the distance between two distributions in the CGAN loss function can effectively avoid the phenomenon of "modal collapse" that may occur in traditional CGAN training due to the disappearance of gradients. According to [30], the Wasserstein distance between the true distribution and the generated distribution can be expressed as:

$$D_w = E_{x\sim P_{data}}[f_w(x)] - E_{x\sim P_G}[f_w(x)] \tag{4}$$

where D_w is the Wasserstein-distance loss. The generator of the CGAN will make the generated distribution close to the true distribution of the data during training, so that the data-generation ability of the

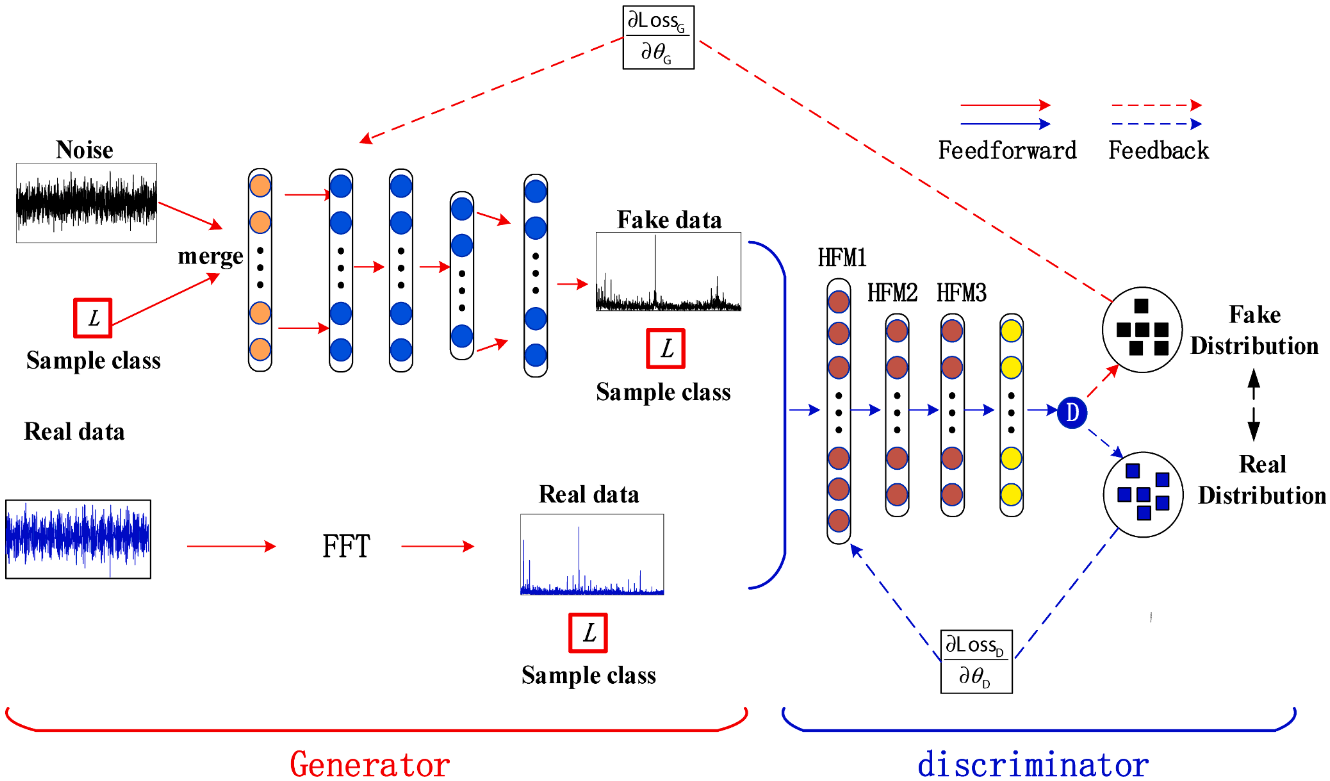

Fig. 4. The structure of the WCGAN-HFM.

generator is continuously enhanced.

3.2. Hierarchical feature matching

The Wasserstein distance can shorten the distance between the generated distribution and the real distribution from a global perspective. For imbalanced sample generation, the approximation of the global distribution forces the network to focus on the normal classes with dominant sample sizes. At the same time, it is difficult to ensure the quality of sample generation by excessively pursuing the approximation of the whole and neglecting the correspondence between classes. Therefore, to ensure the quality of network generation, additional feature-matching loss was added to the network to further constrain the rules of sample generation from the perspective of subclasses in the domain, as shown in Fig. 3.

For a specific layer D_l of the discriminator network, the feature-matching loss can be expressed as the distance between the generated feature distribution and the real feature distribution on D_l. The mean absolute error is selected for the distance characterization because of its high computational efficiency for real industrial real time analysis:

$$fm_l = |D_l(x|y) - D_l(G(z|y))| \tag{5}$$

where fm_l is the feature-matching loss at the l - th layer of the discriminator. For a specific category, Equation (8) aligns the features of the generated and real signals in the latent space, which intuitively constrains the quality of sample generation.

The concept of hierarchy is one of the keys to deep-learning success. For deep networks, different layers can learn different concepts. Therefore, the gap between the generated samples and the real samples can be narrowed from different conceptual perspectives through the hierarchical feature-matching regularization. Then, the overall loss of the hierarchical feature-matching is as follows:

$$FM = \sum_l \omega_l \cdot |D_l(x|y) - D_l(G(z|y))| \tag{6}$$

where ω_l is the weighting factor of the l - th layer loss.

The overall structure of the network is shown in Fig. 4. For the proposed WCGAN-HFM, it first uses unbalanced samples to train the network, so that the network can generate specific class data. Then, pseudosamples are generated to expand the original unbalanced class to weaken the influence of the data imbalance on the network performance. After that, an additional deep classification network is constructed, and the fault classification is realized by expanding the data-set training. The loss of fault classification is accomplished using the traditional cross-entropy loss:

$$loss = -[y log\hat{y} + (1-y)log(1-\hat{y})] \tag{7}$$

where $\hat{y}$ is the predicted label.

When the network is fully trained, the generated data will be used to expand the current unbalanced dataset. At this point, we can use any of the classification models to train it and implement subsequent fault diagnosis. Without loss of generality, in this paper, we choose the traditional 3-layer CNN model, which has the same structure as the discriminator network, except that the number of cells in the last layer becomes 4. In subsequent chapters, the effectiveness of the model will be indirectly judged by the accuracy of the classifier.

4. Experiment

4.1. Experiment setting

4.1.1. Parameters of the WCGAN-HFM

Both the generator and discriminator were composed of three convolution layers. The number of convolution kernels was 64, and the sizes of the convolution kernels were 7, 5, and 3. The leaky rectified linear unit (leaky ReLU) activation function was adopted for the whole network, and batch normalization was used for normalization. For the generator, the last layer of the activation was 'tanh.' For the discriminator, the last layer did not use any activation functions. HFM was

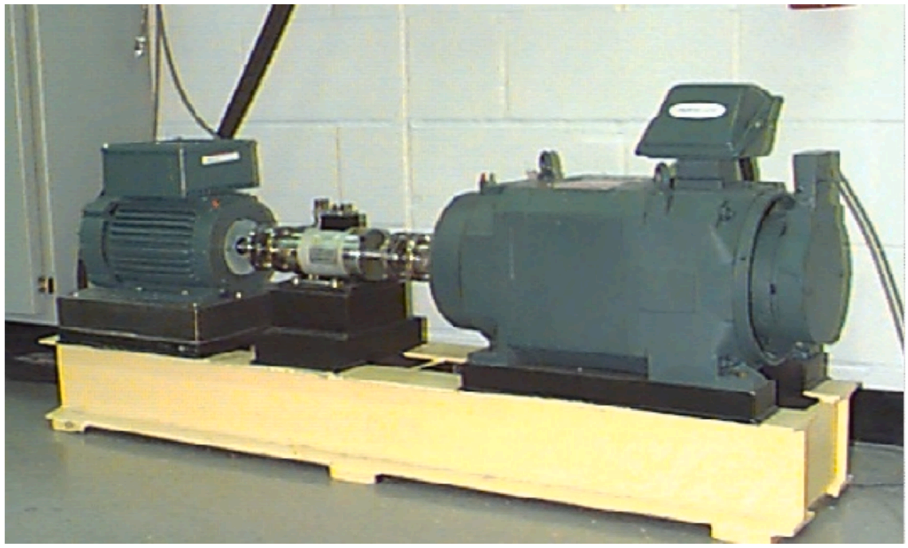

Fig. 5. Test-rig of the rolling bearing.

Table 1
The experiment setting.

	Training Samples	Generated samples	Test samples
Case 1	Normal (500), inner (5), outer (5), roll (5)	Inner (250), outer (250), roll (250)	Normal (1888), inner (3559), outer (3559), roll (3559)
Case 2	Normal (500), inner (10), outer (10), roll (10)	Inner (250), outer (250), roll (250)	Normal (1888), inner (3554), outer (3554), roll (3554)
Case 3	Normal (500), inner (15), outer (15), roll (15)	Inner (250), outer (250), roll (250)	Normal (1888), inner (3549), outer (3549), roll (3549)

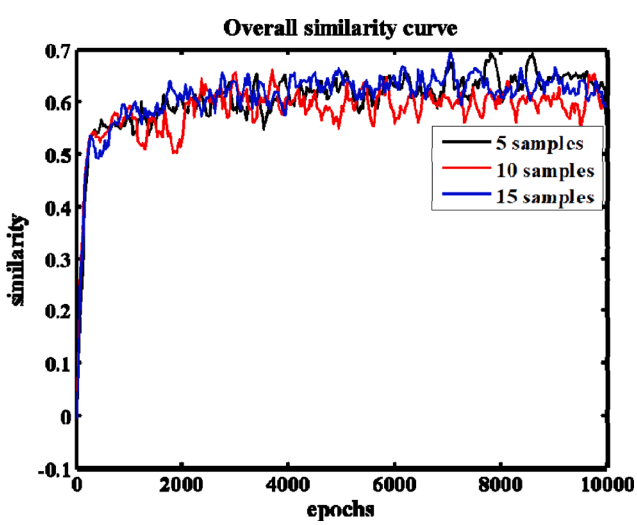

Fig. 6. The curves of similarity change with the training process.

applied to the three layers of the discriminator, with the weights of the layers being 10, 1, and 1, respectively. The structural parameters and hyper-parameters of the network could be obtained by testing the convergence of the network and the quality of the generated samples. The optimizer of the network was RMSprop, and the learning rate was equal to 0.0001. The training epochs were 10000, and the batch size was 32.

The additional classification network consisted of three convolution layers. The number of convolution kernels was 64, and the size of the kernel was 3. The rest of the network configuration was the same as the discriminator, except that the last layer was activated using the 'softmax' function.

To highlight the advantages of the proposed method, the k-nearest neighbors (KNN), support-vector machine (SVM), deep neural network (DNN), convolutional neural network (CNN), CNN-SVM [23], E-GAN [24] and WCGAN methods were selected for comparison. It is worth mentioning that the CGAN was not adopted as a comparison algorithm due to its structure being difficult to be optimized along with its poor performance. The methods used had been fine-tuned experimentally to better suit the task.

4.2. Comparative experiment 1

4.2.1. Data set description

This study used data collected by the Case Western Reserve University Bearing Data Center to verify the proposed method. As shown in Fig. 5, the experimental platform included a two-horsepower motor, a torque sensor, a power meter, and electronic-control equipment. The bearing type tested was 6205-2RS JEM SKF. Faults were processed by electro-discharge at the inner ring, ball, and outer ring, respectively. The motor-load range was 0 to 3 horsepower (the motor speed was 1720 to 1797 r/min), and vibration data were recorded at sampling frequencies of 12 kHz and 48 kHz. 12 kHz collection data are selected for analysis in this paper, and a more detailed description can be found in [31].

To verify the effectiveness of the proposed method to deal with the imbalance problem, we analyzed and verified the network under four kinds of loads. The data set included faults of normal bearings, inner rings, outer rings, and balls under four loading conditions. In particular, each type of failure contained three levels of fault. In this study, all of the data were divided by time windows, each with length of 1200 and a step size of 400. Normal, inner ring, outer ring, and ball faults included 2388 (4 conditions * 597 samples), 3564, 3564, and 3564 (4 conditions * 3 fault levels * 297 samples) samples, respectively. For each type of fault, we selected 5, 10, and 15 fault samples to simulate the unbalanced state, while the number of normal samples was determined to be 500. The detailed experiment setting is shown in Table 1.

4.2.2. Experiment results

Fig. 6 shows the changes in the samples generated by the generator with the training epoch during the training of the three cases. It was found that the generation performance of the generator gradually

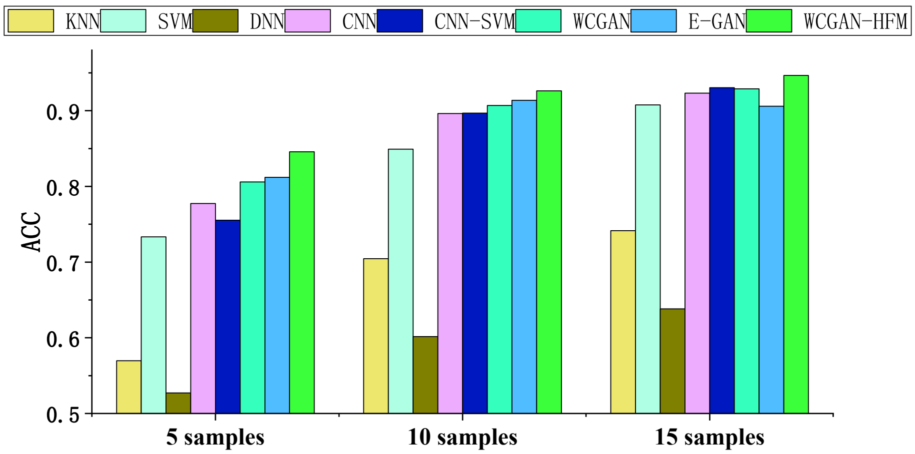

Fig. 7. The diagnostic results of the different methods in three cases.

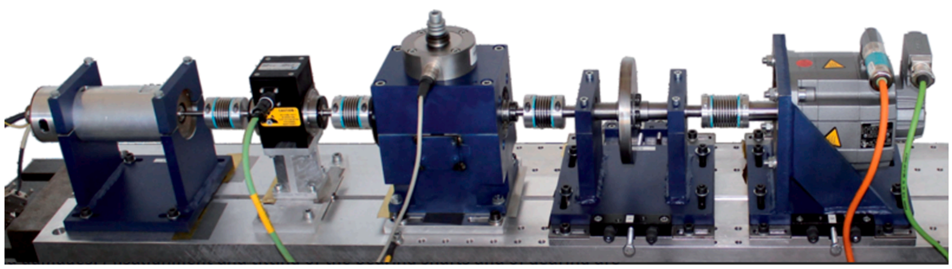

Fig. 8. Test-rig of KAt-DataCenter.

Table 2
The experiment data description.

Normal(Class1)	Inner fault(Class2)	Outer fault(Class3)	Mixed fault(Class4)
K001K002K003	KI04KI14KI16	KA04KA22KA30	KB23KB24KB27

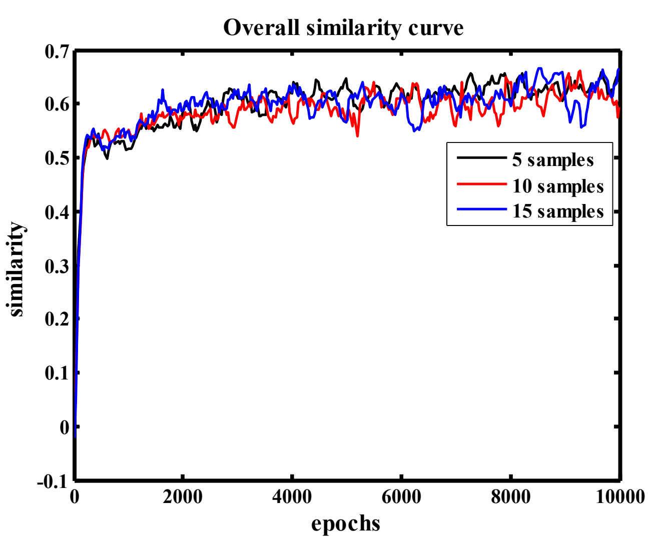

Fig. 9. The similarity curve of KAt-DataCenter dataset.

improved and stabilized in the later period. Owing to the randomness of the noise used in the generation of samples, the similarity curve had certain fluctuations in the later period, which was acceptable. The high similarity ensured the effectiveness of the network training, and the introduced randomness could enhance the robustness of the network in dealing with noise.

Fig. 7 plots the accuracy of the different methods in the three cases. DNN is found to have the worst performance, presumably due to too few samples and severe overfitting of the training data. Similar problems exist for KNNs, whose generalization ability is also positively correlated with the data. For support vector machine, it uses support vectors to delineate decision boundaries, so it has a good performance similar to CNN in the imbalance problem. CNN-SVM is a fusion of the two algorithms, but its performance is unstable and only has performance improvement in case 3. E-GAN also uses generated data, but similar to WCGAN, and they only regularize the overall distribution of the data. This reduces the quality of data generation, which in turn affects the diagnostic accuracy. The proposed WCGAN-HFM uses Wasserstein loss to stabilize the training process on the basis of the original CGAN. On this basis, the features of the generated samples are further regularized through hierarchical feature matching to obtain the highest accuracy.

4.3. Comparative experiment 2

4.3.1. Data set description

In this section, the actual fault bearing data from KAt-DataCenter [32] is used. This data set was established by Christian Lessmeier, Chair of Design and Drive Technology at Paderborn University [33]. The difference from CWRU is that the bearing fault of this data set are obtained through the life-cycle experiment, so it is closer to the actual industrial working conditions. The test rig is shown in Fig. 8. The bearing type is 6203, the sampling rate is 64khz, and the sampling time is 4 s. This paper selects three normal bearings, three outer fault bearings, three inner fault bearings, and three mixed fault bearings (inner fault + outer fault) for research. The specific parameters are shown in Table 2.

Similarly, the move-time window is used for data partitioning. The window length is 1200, and the moving step is 400. It is worth noting

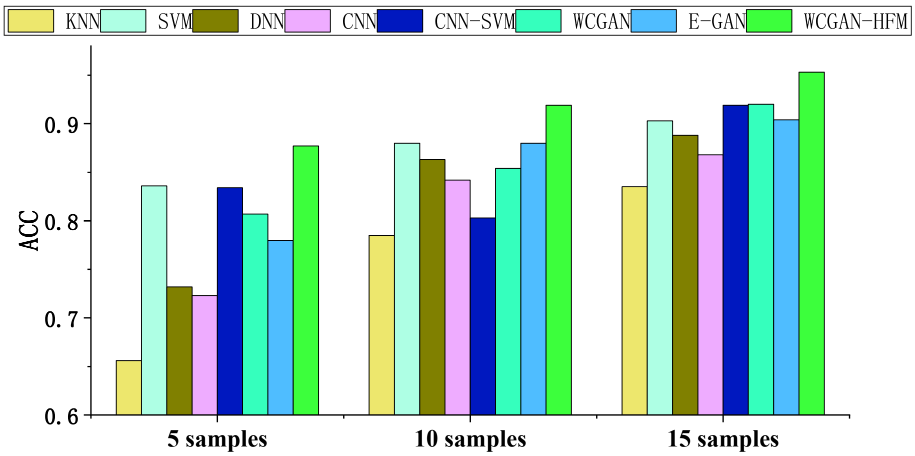

Fig. 10. The diagnostic results of the different methods in KAt-DataCenter dataset.

that in order to increase the amount of information in the data within a window, we performed a frequency reduction operation on the data. The number of samples for normal, inner, outer and mixed faults after data division is 1827. The experimental setup is also consistent with CWRU, where 5, 10 and 15 samples are used for training and the rest for testing.

4.3.2. Experiment results

From Fig. 9, it can be found that WCGAN-HFM still converges relatively stably, which proves the adaptive performance of the proposed method. Fig. 10 shows the diagnostic results on real bearing faults, and it can be found that the proposed method still achieves the optimal results. It is noteworthy that there are some findings that are inconsistent with those of the CWRU dataset. The DNN slightly outperforms the CNN on the KAt-DataCenter dataset. Meanwhile, it can be found that the performance of CNN-SVM is not monotonically increasing with the increase of samples. We speculate that the occurrence of these phenomena depends on the representativeness of the samples used.

In this dataset, it can be found that the gap between the proposed method and the compared GAN-related methods is more pronounced. We speculate that this is due to the fact that the real bearing fault signals are contaminated with noise, the signal-to-noise ratio of the signals is lower, and therefore higher demands are placed on the quality of the generated data. This phenomenon highlights the efficacy of the proposed hierarchical feature matching and proves the superiority of the proposed method.

4.4. Visualization results

To highlight the superiority of the proposed algorithm in the generation of unbalanced samples and the effectiveness of the data-expansion strategy, this section shows samples of different types of faults generated by the network. Additionally, the characteristics of the generated samples and the real samples in the deep layers of the classification network were also visually analyzed.

Fig. 11 shows the true frequency spectrum of the four different states of the bearing, the frequency spectrum of the data generated by the WCGAN, and the frequency spectrum generated by the WCGAN-HFM. It could be found that the proposed method adopted extra regularization, which prompted the generator to generate more realistic samples. Therefore, the generated signal was similar to the real signal. In contrast, the samples generated by the WCGAN were heavily affected by noise. The visualization results highlighted the effectiveness and superiority of the proposed method in data generation.

To visually analyze the effectiveness of the expanded sample on the classification task, the features of the penultimate layer of the classification network were visualized using t-distributed stochastic neighbor embedding (t-SNE). As shown in Fig. 12, the test data were assigned four colors of cross-shaped representation, and the three types of generated fault data were marked with a rice shape. It is worth noting that in the KAt-DataCenter dataset, the label of roll refers to a mixed fault. From Fig. 12, it can be found that the test data are more divergent on the two datasets, which is because the classification network has never been exposed to the test data. For example, in Fig. 12 (a), the specific fault classes become three different aggregations in the corresponding classification space, which correspond to the three bearing individuals in each class. The generated data is exactly in the middle of all the data, filling the gap in the classification space. The phenomenon in Fig. 12(b) is similar to that in Fig. 12(a), i.e., the test data are clustered into three clusters in their respective classification spaces, and the generated data are all very close to the tested data. This proves the reliability of the data generated by the proposed method.

5. Conclusions

In this paper, a new unbalanced sample fault diagnosis model, WCGAN-HFM, is proposed to solve the problem of data unbalance in real industrial environments. The model introduces Wasserstein distance loss on the basis of traditional CGAN to solve the problems of model training instability and modal collapse. Meanwhile, hierarchical feature matching loss is proposed to further standardize the quality of the samples generated by the model. Different from existing generative models, the proposed WCGAN-HFM regularizes the generated data from the global perspective and the perspective of intra-class within the distribution to achieve a multi-angle approximation of the generated samples to the real samples and enhance the validity of the generated data. The experimental results prove the superiority of the proposed method, and the following three conclusions can be drawn.

1) The results of bearing data from CWRU and real bearing fault data from KAt-DataCenter prove the effectiveness and generalization performance of the proposed method. With only a very small number of 5 or 10 samples, the proposed method can achieve to obtain a high fault diagnosis rate.
2) The generated data visualization results show that the proposed WCGAN-HFM is able to generate meaningful new fault signals matching the faults, proving the validity and reliability of the proposed method.
3) The results of the comparison experiments show that the proposed method outperforms state-of-the-art methods, which proves the superiority of the proposed method.

The proposed method has been tried only in the case of stable

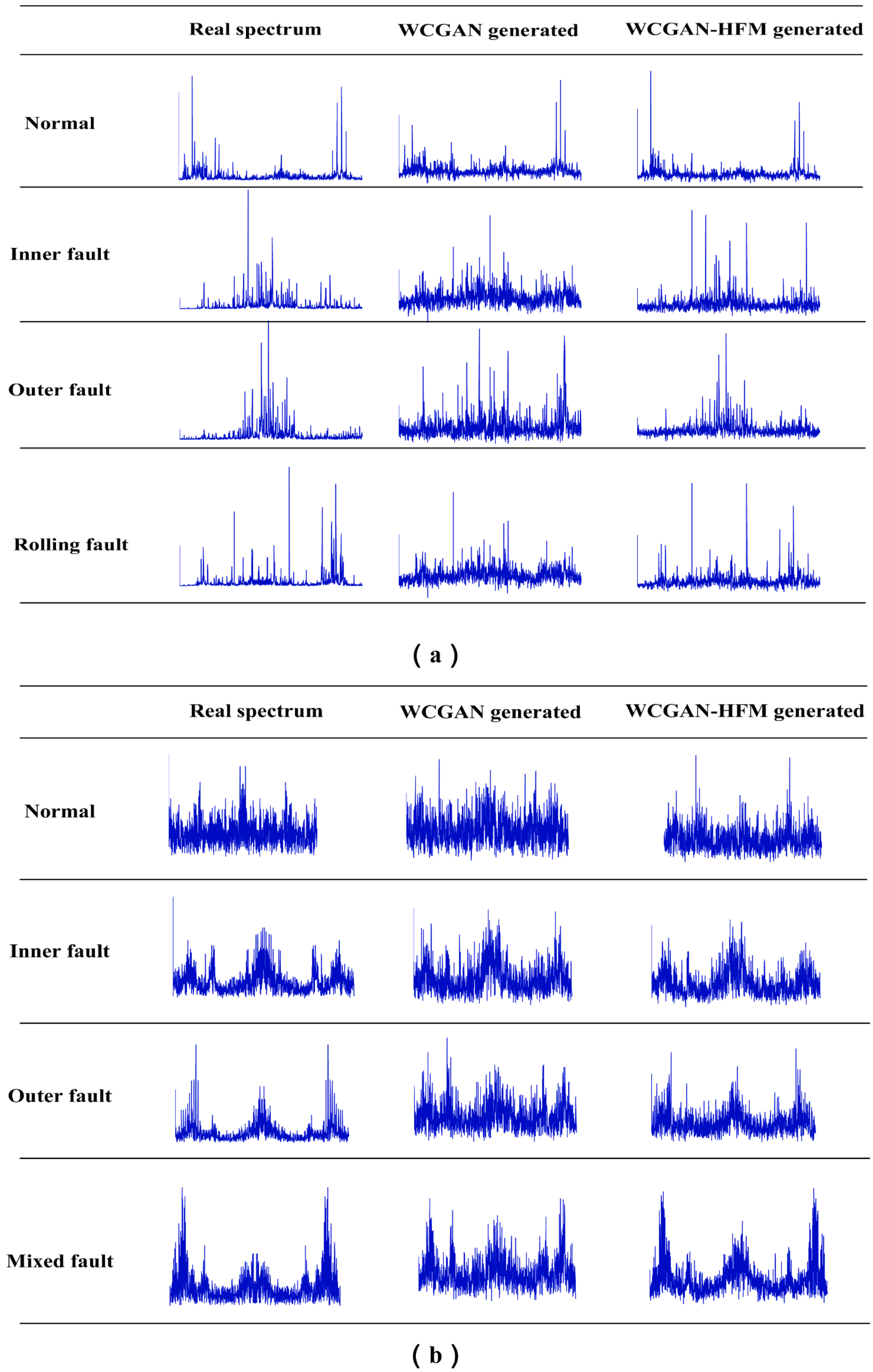

Fig. 11. The visualization of real and generated signals. (a) CWRU dataset; (b) KAt-DataCenter dataset.

rotational speed, however, there may be fluctuations in the working conditions of the actual industry. How to further improve the generalizability and robustness of the proposed method will be the focus of the next research.

CRediT authorship contribution statement

Yizhen Peng: Conceptualization, Methodology, Software, Writing – original draft, Funding acquisition. **Yu Wang:** Visualization, Investigation, Writing – review & editing. **Yimin Shao:** Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

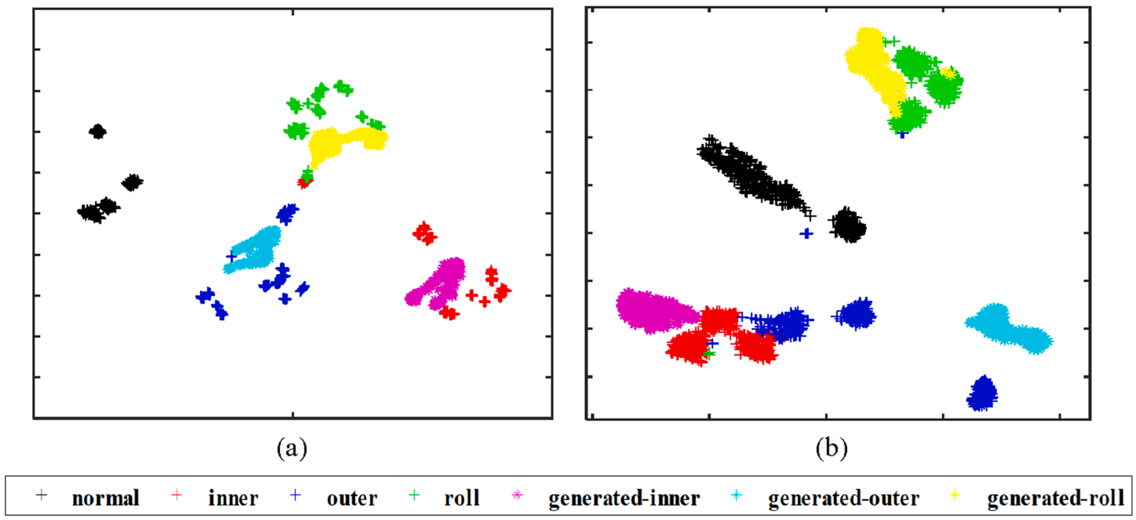

Fig. 12. The visualization results. (a) CWRU dataset; (b) KAt-DataCenter dataset.

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